

$$\text{error}(t) = e(t)$$

$$e(t) = \text{target position} - \text{measured position}$$

$$\text{control signal} = u(t)$$

$$u(t) = \underbrace{K_p \cdot e(t)}_{\text{Proportional}} + \underbrace{K_i \cdot \int e(t) \cdot dt}_{\text{integral}} + \underbrace{K_d \cdot \frac{de}{dt}}_{\text{derivative}}$$

3 constants

- K_p
- K_i
- K_d

Proportional

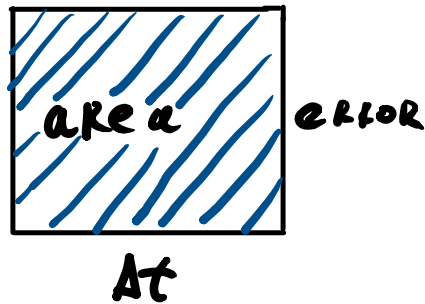
$$K_p \cdot e(t)$$

No changes needed for microcontroller

integral

$$\int e(t) dt \approx e_{int} + e \cdot \Delta t$$

$e \cdot \Delta t$ is area of rectangle



e_{int} is the accumulated integral value from the previous steps

SO: $e_{int} = \text{Previous accumulated area}$

So $I_{term}(k-1) + K_i \cdot e(t) \cdot \Delta t$

if $\Delta t \rightarrow 0$ then discrete sum $\rightarrow$ TRUE integral.

Every loop

- Measure the error
- Multiply by Δt
- add the stored integral memory
- Memory slowly grows

if error = 5

if $ERROR = 5$

$\Delta t = 0.01$

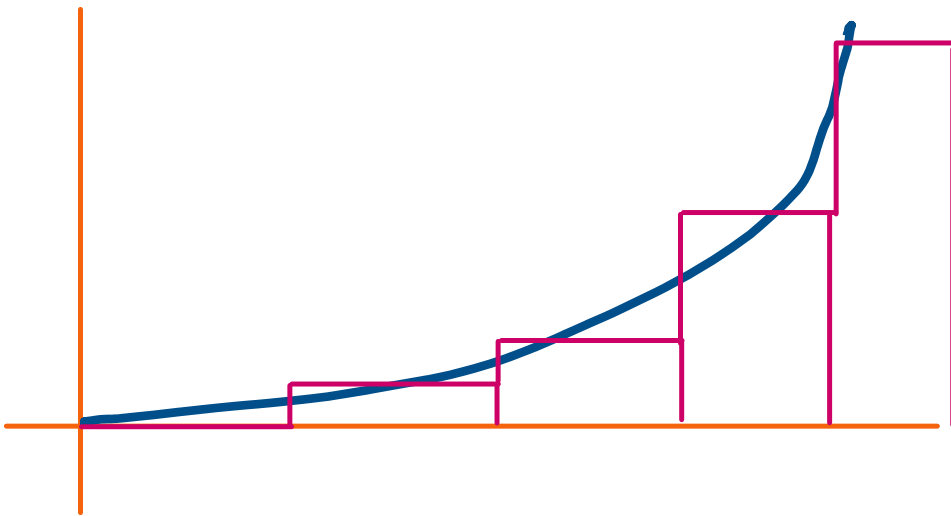
each loop adds: $5 \times 0.01 = 0.05$

after 100 loops $Integral = 100 \times 0.05 = 5$

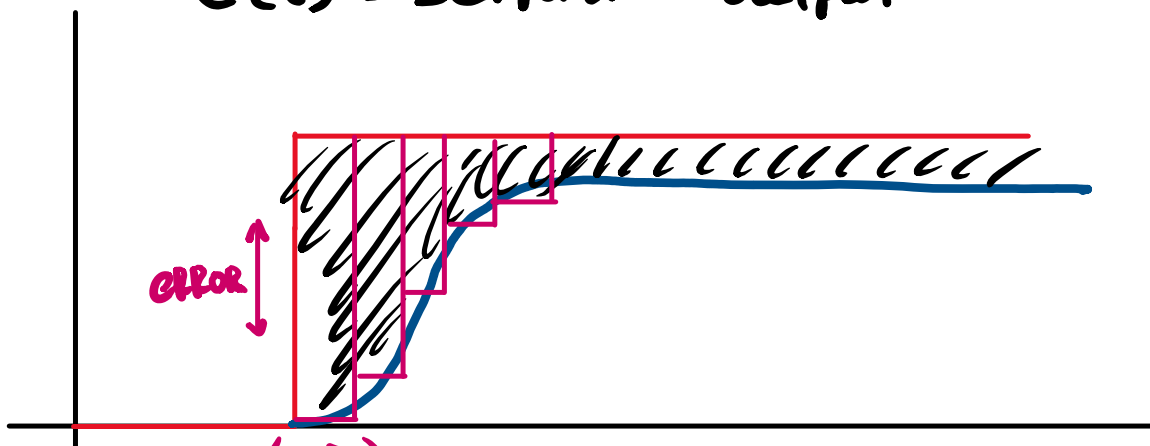
Same as

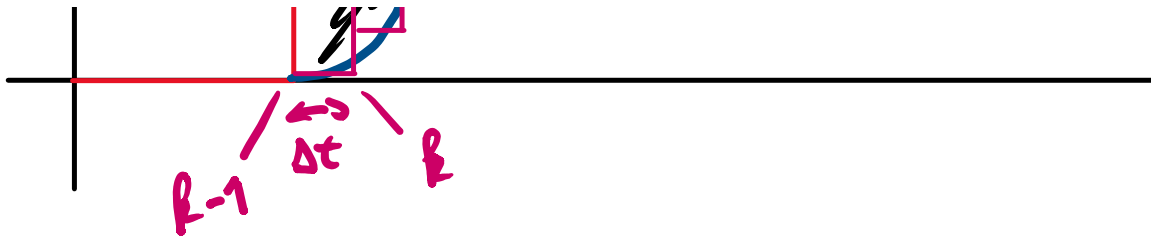
$$\int 5 dt = 5t$$

if $ERROR$ is negative
then subtracting Rectangles



$$e(t) = \text{Setpoint} - \text{Output}$$

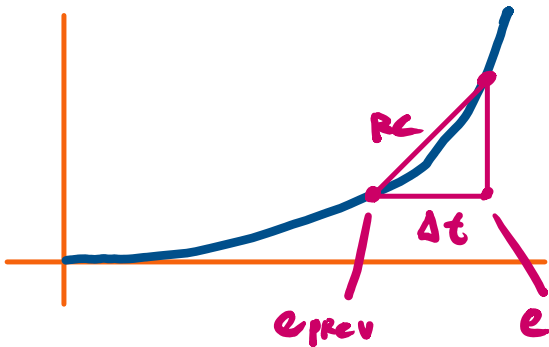




So changes needed for microcontroller

Derivative

$$\frac{de}{dt} \approx \frac{e - e_{prev}}{\Delta t}$$



Δt is same as T_s the sampling time.

Changes needed for microcontroller

Microcontroller

$$u(t) = K_p \cdot e(t) + K_i \cdot \int e(t) \cdot dt + K_d \cdot \frac{de}{dt}$$



$$u(t) = K_p \cdot e(t) + I_{term} + K_i \cdot e \cdot \Delta t + K_d \cdot \frac{e - e_{prev}}{\Delta t}$$

for Drone

$$e(t) \Rightarrow e(k) = \text{Desired}(k) - \text{Measure Rate}(k)$$

iteration number

$$e_{\text{prev}} \Rightarrow e(k-1)$$

$$I_{\text{term}} \Rightarrow \text{PrevIterm} = I_{\text{term}}(k-1)$$